

Date	01 October 2026	Team ID	SWTID-2026-2668
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

DATA FLOW DIAGRAMS:

A **Data Flow Diagram (DFD)** is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Operational Flow: (ServiceNow Laptop Procurement System)

- User configures hardware variables (RAM, SSD) for the Standard Laptop (Lenovo ThinkPad X1 Carbon) in Service Portal and submits the requisition cart (\$1,100).
- Service Catalog engine creates parent Request ('sc_request') and Requested Item ('sc_req_item') records, and routes an authorization document in 'sysapproval_approver' to the Line Manager.
- Line Manager reviews order justification and spend commitment, authorizing the order to transition state from 'Requested' to 'Approved'.
- Approval state update triggers the Flow Designer workflow 'Standard laptop task' executing under the elevated 'System User' context in the 'Global' application scope.
- Flow action 'Create Catalog Task' generates child task 'sc_task' pre-populated with Short Description 'Laptop need to Configured' assigned to the Hardware fulfillment queue.

DFD LEVEL 0: STANDARD LAPTOP PROCUREMENT CONTEXT DIAGRAM

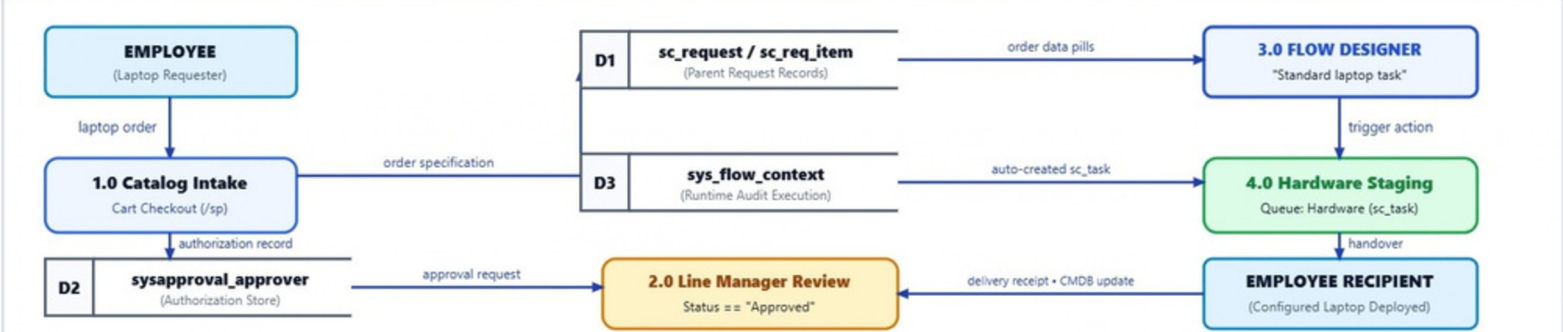


FIGURE 1: DFD LEVEL 0 — STANDARD LAPTOP PROCUREMENT FLOW DESIGNER ARCHITECTURE

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web user)	Service Catalog Intake	USN-1	As an employee, I can view and select the Standard Laptop item (Lenovo ThinkPad X1 Carbon).	Item is visible under Hardware category with technical specs and \$1,100 fixed price.	High	Sprint-1
Customer (Web user)	Service Catalog Intake	USN-2	As an employee, I can configure hardware options (RAM, SSD) and enter business justification.	Form validates inputs and prevents checkout if mandatory justification is blank.	High	Sprint-1
Customer (Mobile user)	Mobile Portal Intake	USN-3	As a mobile employee, I can submit the laptop order using the ServiceNow Now Mobile application.	Mobile interface captures specifications and creates REQ/RITM records cleanly.	Medium	Sprint-1
Customer (Web user)	Order Tracking	USN-4	As an employee, I can track procurement progress through parent REQ and RITM reference numbers.	Order status reflects real-time stages (Requested → Approved → Staging).	Medium	Sprint-1
Customer (Web user)	Order Notifications	USN-5	As an employee, I will receive email notifications once my laptop order is submitted and approved.	System dispatches notification containing requisition ID and expected delivery SLA.	High	Sprint-1
Customer (Approver)	Approval Governance	USN-6	As a manager, I can review pending laptop orders and approve expenditure in the portal.	Manager opens sysapprova1_approver record and sets state to "Approved".	High	Sprint-1
Customer (Approver)	Approval Governance	USN-7	As a manager, I can reject an order with feedback comments if equipment is unjustified.	System cancels RITM, notifies employee, and halts task creation cleanly.	Medium	Sprint-1
Administrator	Flow Designer Automation	USN-8	As an administrator, I can configure the flow trigger on Service Catalog submissions.	Flow triggers automatically on submission and evaluates manager authorization.	High	Sprint-2
Administrator	Flow Security Scope	USN-9	As an administrator, I can configure the flow to execute as "System User" in Global scope.	Sub-flow executes with elevated background system privileges without caller ACL errors.	High	Sprint-2
Administrator	Process Engine Binding	USN-10	As an administrator, I can bind the flow via Maintain Items Process Engine.	Process Engine tab binds to Standard Laptop task, deprecating legacy plans.	High	Sprint-2
Customer Care (Hardware Tech)	Hardware Staging	USN-11	As a technician, I receive auto-created tasks: Short Desc "Laptop need to Configured".	Task appears in Hardware queue with state "Open" within 2 seconds of approval.	High	Sprint-2
Customer Care (Hardware Tech)	Staging & CMDB Sync	USN-12	As a technician, I record asset serial numbers, complete imaging, and close task.	Task marks "Closed Complete" and device record in a1m hardware updates to "In Use".	High	Sprint-2
Customer Care (Service Desk)	Audit & Oversight	USN-13	As a coordinator, I can review automated flow execution traces in 'sys_flow_context'.	Execution history confirms automated task creation with 0 manual coordinator handoffs.	Medium	Sprint-2